

Zelong Xu

EDUCATIONAL BACKGROUND

University of Wisconsin-Madison

Sep 2024 – Dec 2026

B.S. in Statistics (expected Dec 2026)

Certificates: Computer Science, Mathematics

- GPA: 3.90/4.00
- Core Courses: Interpretable Machine Learning, Matrix Methods in Machine Learning, Deep Learning, Numerical Analysis

Shandong University (transferred to UW-Madison)

Sep 2022 – Jun 2024

- GPA: 88.1/100

PUBLICATIONS

DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment

Jan 2026 – May 2026

EMNLP 2026 · Advisor: Prof. Yue Zhao · [arXiv]

- Proposed DOG-DPO, a training-free data selection framework that represents each preference pair as a direction in representation space and decomposes multi-dataset preference geometry into a global anchor subspace and dataset-specific residual subspaces, selecting subsets by maximizing diversity-based coverage of alignment directions.
- Using only 11.6% of preference pairs, recovers most safety gains of full-data DPO across six safety benchmarks and two backbones (LLaMA-3.2-3B, Gemma-3-1B), while remaining teacher-free and $15\text{--}35\times$ faster than representative selection baselines (DEITA, Selective-DPO).

PREPRINTS

Statistical Confidence for LLM Circuit Discovery via Randomized Interventions

May 2026 – Sep 2026

Under review · First author · Advisor: Prof. Yang Lu

- Introduced a randomized-intervention framework that estimates each edge’s average causal contribution across circuit densities (sparse to near-complete) and tests it with calibrated p-values, reusing the same model evaluations for all candidate edges so that multiple-testing correction remains practical at scale.
- Across five model–task settings (up to 34,000+ candidate edges), p-values stay uniform or conservative on structural negative controls with zero false discoveries after correction; fixed-circuit analyses further reveal superadditivity — combining variable with shared components more than doubles recovery.

CUA-SWE: When Computer-Use Agents Meet Visual Software Engineering

Jun 2026 – Sep 2026

Under review · Co-first author · Advisors: Prof. Yue Zhao and Prof. Xiyang Hu

- Co-developed CUA-SWE, a benchmark, interactive environment, and evaluation pipeline for software engineering with computer-use agents, spanning web, game, mobile, and DevOps domains; each task pairs an editable project with running software that agents operate and visually inspect, verified by deterministic task-specific tests.
- Evaluated frontier agents under code-only and hybrid CUA conditions, showing GUI access raises task success by up to +48.6 pp and characterizing how agents use visual feedback to diagnose, repair, and verify software changes.

SyncRA: Learning Temporal Correspondence in Omni-Modal Models

Apr 2026 – Sep 2026

Under review · First author · Advisor: Prof. Xiyang Hu

- Proposed SyncRA, a lightweight method that strengthens audio–visual temporal correspondence in omni-modal models by contrasting intermediate audio–visual representations within each video (symmetric InfoNCE over native temporal intervals), deriving supervision from input timing alone with no extra annotations and no inference-time cost.
- Designed a controlled temporal-swap diagnostic revealing that models answer as if original pairings persist; across four open omni-modal backbones and five video benchmarks, SyncRA consistently outperforms answer-only fine-tuning.

OTHER RESEARCH EXPERIENCE

EmoPair: Counterfactual Paired Listwise Alignment for Emotion-Aware Speech LLMs *Nov 2025 – May 2026*

- Proposed EmoPair, a counterfactual paired listwise alignment framework for emotion-aware speech LLMs that synthesizes same-text-multiple-emotion speech variants (7 Ekman emotions) via Qwen3-TTS and trains the model with listwise preference optimization plus a paired counterfactual constraint anchoring emotional responses against a neutral baseline.
- Outperformed binary-GRPO (PALLM) and Likert-GRPO (ParaS2S) baselines on the same Qwen3-Omni-30B backbone, improving response appropriateness from 72.4 to 77.2 on IEMOCAP and 47.9 to 55.8 on held-out RAVDESS (OOD), and raising counterfactual swap consistency from 0.62 to 0.81.

BR-LOCAL: A Solver-Verified Benchmark for Ripple Effects under In-Context Knowledge Editing *Oct 2025 – April 2026*

- Designed BR-LOCAL, the first solver-verified benchmark for ripple-effect evaluation under in-context knowledge editing (ICE), addressing the 33–76% annotation error in Wikidata-grounded predecessors (MQuAKE, RippleEdits): minimal patches on synthetic Horn knowledge bases yield episodes with deterministic gold change-sets and continuous per-query distance-to-patch $d(q)$, including an unreachable ∞ -stratum; paired COLD-START and ONLINE-FULL tracks isolate static reasoning from in-context edit propagation.
- Built a procedural Horn-KB synthesizer and end-to-end evaluator harness producing 5K episodes; benchmarked SOTA LLMs like GPT-5.2, DeepSeek-R1, Claude-4.6, and Qwen3-235B, activating the reasoning chain raises Change-F1 from 0.12 to 0.90, establishing that ripple-effect capability is gated by inference-time reasoning rather than parametric knowledge.

Multi-Emotional Probing for Controllable Language Generation *Mar 2025 – May 2025*

- Proposed and implemented a novel probe-and-inject framework to achieve fine-grained emotional control in LLMs, successfully extracting emotion vectors from BERT’s latent space and steering Llama-2 model’s generative output.
- Discovered that emotion representations are localized in BERT’s middle layers (5-17) and exhibit a near-orthogonal geometric structure, enabling precise control with high F1-scores for targeted emotions like ‘joy’ (0.93) and ‘anger’ (0.90) via vector injection.

INTERNSHIP EXPERIENCE

Icarus Fund, Remote *Jul 2024 – Aug 2024*
Quantitative Analyst

- Developed and backtested Python-based quantitative models for S&P 500 companies, optimizing portfolio strategies with Sharpe Ratios up to 1.42.

SKILLS & LANGUAGES

Technical Skills: Python, Java, JavaScript, HTML, R, SQL, PyTorch, TensorFlow, Scikit-learn, SPSS, Stata

Languages: Chinese (Native), English (Fluent, TOEFL: 109/120)